Real Time Air Quality Analytics of Chennai and its Implications on Health

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Abstract
Monitoring and classifying air quality is an intricate task due to the vibrant nature, instability, and high inconsistency in time and space of pollutants and particulates. At the same time, being able to model, forecast, and observe air quality is becoming significant, specifically in metropolitan areas due to the witnessed dangerous impact of air pollution on residents' well-being and the environment. To analyse the AQI in Chennai city, the data was collected from 2019 to 2021 for the areas Kodungaiyur, Perungudi and Royapuram, then pre-processed to replace missing values and remove redundant data. The combination of Random Forest, K-Nearest Neighbour (KNN) and Decision Tree Model was used to classify the AQI values. The Random Forest method achieves higher accuracy of 100 per cent compared to the KNN and Decision Tree Model. Analysis of AQI values can control pollution levels by incorporating various strategies and policies by the government.

Keywords: Air Quality Index (AQI), Random Forest, Decision Tree, KNN, Pre-processing

Introduction
With the economic and technical growth of cities, environmental pollution complications are arising, such as water pollution, noise pollution , and air pollution. In particular, air pollution has a direct impact on human health through the exposure of pollutants and particulates, which has increased interest in air pollution and its impacts among the scientific community (Hvidtfeldt, 2018; Gonzalez, 2017; Pimpin, 2018). The main causes associated with air pollution are the burning of fossil fuels, agriculture, exhaust from factories and industries.

Air quality monitoring serves several purposes. Firstly, it involves monitoring the main pollutants present in the air environment on a regular or continuous basis to determine whether the air quality meets the national standards and to provide data for air quality evaluation reports. Secondly, it provides a basis for studying the patterns and trends in air quality and forecasting air pollution. Lastly, it provides the necessary information for government departments to implement relevant environmental protection laws and regulations, carry out environmental quality management, conduct environmental research, and revise atmospheric environmental quality standards.

The Environmental Protection Agency (EPA) tracks the commonly known criteria pollutants i.e. ground-level ozone (O_3), sulphur dioxide (SO_2), particulate matter (PM10 and PM2.5), carbon monoxide (CO), carbon dioxide (CO_2), and nitrogen dioxide (NO_2). These substances are in compositions of a common index, called the Air Quality Index (AQI), indicating how clean or polluted the air is currently or forecast to become in specific areas. As the AQI increases, a higher percentage of the population is exposed.

In the face of increasingly serious environmental pollution problems, scholars have conducted a significant quantity of related research, and in those studies, understanding the causes of air pollution has been of paramount importance. Thus, with full knowledge of the increasing pollution derived problems, the importance of accurately forecasting the levels of air pollutants has increased, playing a critical role in air quality management and population prevention against pollution hexes.

In India, the Central and State Pollution Control Boards have commissioned the National Air Monitoring Programme (NAMP) which covers 240 cities with 342 monitoring stations. Air Quality Index (AQI) has been categorised into different groups. AQI is measured by numerical values from 0 to 500. When the values are 0, it is the normal adequate air quality, and 500 (high AQI pollution) is the most hazardous air quality. AQI is classified into six groups centred on the environmental abundance and health impacts of air pollution (identified as safety interruptions). These six grades are the results: Good (safe), Moderate, Sensitive, Unhealthy, Very Unhealthy, Dangerous. The economy, food development, transport and its use are human society's practices that influence the climate. Air quality is affected by these activities, either directly or indirectly.

To analyse the AQI in Chennai city, the data was collected from 2019 to 2021 for the areas Kodungaiyur, Perungudi and Royapuram, then pre-processed by replacing missing values and removing redundant data. The combination of Random Forest, KNN and Decision Tree Model is used to classify the AQI values. Analysis of the AQI value can control pollution levels by incorporating various strategies and policies by the government.

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environmental protection laws and regulations, carry out environmental quality management, conduct environmental research, and revise atmospheric environmental quality standards.

The objective of the work is to emphasise the importance of monitoring and classifying air quality in urban areas due to the negative impact of air pollution on public health and the environment. The work focuses on a specific case study in Chennai, where data was collected and pre-processed to analyse the Air Quality Index (AQI) values in three different areas. The text also discusses the use of three different models (Random Forest, KNN, and Decision Tree) to classify AQI values, with the Random Forest method achieving the highest accuracy. Finally, the analysis of AQI values can inform policies and strategies to control pollution levels, highlighting the potential benefits of accurate air quality monitoring and classification.

Review Study

Many literatures have used different statistical methods to study air quality models. Linear regression analysis models are usually used to discuss air quality issues. The advantages of this model are simple calculation, easy interpretation of regression coefficients, and unique output results, but it is not suitable for dealing with non-linear problems. The Markov model is often used to predict the concentration of pollutants in the air (Denglong , 2021).

This method has a good effect on the state of the process, but it is not suitable for medium- and long-term prediction of the system. Some scholars use random forest algorithm 14 to build air quality prediction models, but random forests have been proven to overfit in some noisy classification or regression problems. There are also some literatures using artificial neural networks to establish air quality prediction models. The artificial neural network turns the characteristics of all problems into numbers and turns all reasoning into numerical calculations, and is unable to explain its own reasoning process and reasoning basis.

The Autoregressive Integrated Moving Average Model (ARIMA) is one of the most important and widely used models to forecast time series. It achieved high popularity due to its statistical properties, adaptability to represent a wide range of processes, and the ability to be extended through the years. Ever since the issues of air quality and quality of life in urban areas have raised concern, statistical methods like ARIMA have been widely used to forecast the levels of air pollutants and air quality (Kulkarni, 2018).

With the increasing amount of historical data available for analysis and the need for performing more accurate forecasts in different scientific areas and domains, Machine Learning (ML) models have drawn attention, establishing themselves as a solution that can replace the more classical statistical models in time-series forecasting. Specifically, ML algorithms have been widely used to forecast air quality. The forecast of results of mass concentrations of population and built-up areas has a crucial role to play in taking decisions on atmospheric resources. Such scenarios result in adverse health effects, such as excessive cardiovascular and respiratory disease morbidity and mortality. It is therefore crucial to avoid air pollution in advance by improving air quality protection and ensuring effective environmental monitoring. This is very critical for people's everyday safety and reason for the government to decide on air quality regulation (Liang, 2020).

The two main types of air pollution are outdoor and indoor pollution(Lakshmi, 2013). In this work, the author analyses pollution data from government sources and utilises artificial intelligence tools to predict future levels of pollution. Specifically, the paper models and predicts the levels of carbon monoxide in the atmosphere, primarily caused by vehicular emissions as a major source of outdoor pollution.

The study is based on data from past studies compiled in the virtual repository IndAIR (Pandya, 2022) to guage the development of air quality management policies in India. Additionally, it utilises GIS to map the spatial distribution of research gaps across the country. Through gap analysis, the research reveals that air pollution has been extensively studied in the Indo-Gangetic Plain (IGP) and metro cities, but there are still gaps in knowledge in other areas.

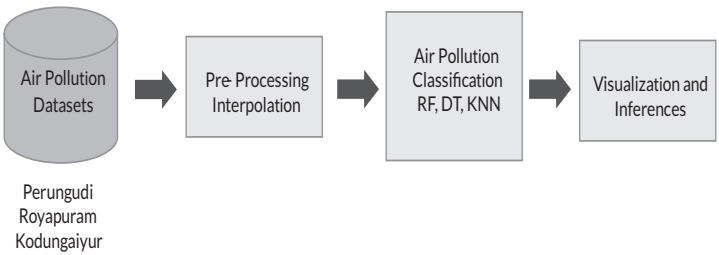
In today's world, monitoring and predicting air quality have become increasingly vital, especially in developing nations such as India (Kumar, 2022). Machine Learning techniques have proven to be the most effective tools for studying modern hazards, in contrast to traditional methods. This study focuses on analysing and predicting air quality based on six years of air pollution data from 23 Indian cities.

It is very important to measure air quality and use the results to predict and detect the relationships between different urban problems. Air quality has influenced urban sustainability. The data on air quality in real-time, which includes data on ozones, particulates, carbon monoxide, sulphur dioxide and nitrous oxide, is vital for regulating air pollution and shielding citizens from harm to their health.

Methodology

The framework of the proposed model is shown in Figure 1. The datasets are first pre-processed followed by the classification phase.

Figure 1: Air Pollution Classification Framework



Source: Authors

Datasets

The datasets have been collected from the Tamil Nadu Pollution Control Board (TNPCB) for the years 2019 to 2021 covering the areas of Kodungaiyur, Perungudi and Royapuram. The air pollution parameters taken for the study were PM_{10}, $PM_{2.5}$, SO_2, NO_2, CO, NH_3, and O_3.

Pre-processing

The datasets collected from the three stations for three years contained missed values so the real-time collection of air pollution sensor data might contain missed values as well.. The datasets are pre-processed by the interpolation technique. Interpolation is a technique used to estimate unknown data points between two known data points. Interpolation is mostly used to impute missing values in the data frame or series while pre-processing data.

Interpolation (Chen, 2021) is mostly used while working with time-series data because here we like to fill missing values with previous one or two values. Linear Interpolation simply means to estimate a missing value by connecting dots in a straight line in increasing order. In short, it estimates the unknown value in the same increasing order from previous values. The default method used by Interpolation is linear. Linear Interpolation is a method for estimating a value that falls between two known data points. It works by assuming that the value between these two points varies linearly with distance. The steps involved in the Linear Interpolation algorithm are as follows:

Algorithm 1. Linear Interpolation

- Step 1: Take two known data points (x_1, y_1) and (x_2, y_2) and the value of an unknown data point x that falls between them.
- Step 2: Calculate the slope of the line that connects the two known points using the formula: slope = $(y_2 - y_1) / (x_2 - x_1)$.
- Step 3: Use the slope to find the value of the unknown data point using the formula: $y = y_1 + \text{slope} * (x - x_1)$.

To interpolate the value of dependent variable y at some point of the independent variable x using Linear Interpolation, we take two points i.e. if we need to interpolate y corresponding to x which lies between x0 and x1 then we take two points (x0, y0) and (x1, y1) and construct Linear Interpolants which is the straight line between these points.

$$y - y_0 = ((y_1 - y_0)/(x_1 - x_0)) * (x - x_0) \qquad (1)$$

And if the value of y is needed to be obtained then using the above equation we calculate yp at x = xp as:

$$y_p = y_0 + ((y_1 - y_0)/(x_1 - x_0)) * (x_p - x_0) \qquad (2)$$

We repeat the same procedure for other points.

Classification

After the pre-processing stage, the air pollution datasets are classified based on the AQI into Good, Satisfactory, Moderate, Poor, Very Poor and Severe. The ranges of AQI for a few pollutants are presented in Table 1. For classification, the RF (Random Forest), KNN (K-Nearest Neighbour) and DT (Decision Tree) machine learning methods are employed in this work.

Table 1: AQI Ranges

AQI Category (Range)	PM10 (24hr)	PM2.5 (24hr)	NO2 (24hr)	SO2 (24hr)
Good (0-50)	0-50	0-30	0-40	0-40
Satisfactory (51-100)	51-100	31-60	41-80	41-80
Moderately polluted (101-200)	101-250	61-90	81-180	81-380
Poor (201-300)	251-350	91-120	181-280	381-800
Very poor (301-400)	351-430	121-250	281-400	801-1600
Severe (401-500)	430+	250+	400+	1600+

Source: Central Pollution Control Bureau (https://cpcb.nic.in/live-air-quality-index/)

Random Forest (RF)

A prominent machine learning method, Random Forest (Singh, 2021), a supervised learning ensemble algorithm, combines multiple decision trees to form a forest and the bagging concept, that later adds the randomness into building the model. The random selection of features is used to split the individual tree while the random selection of instances is used to create a training data subset for each decision tree.

At each decision node in every tree, the variable from the random number of features is considered for the best split. If the target attribute is categorical, Random Forest will choose the most frequent as its prediction. On the other hand, if it is numerical, the average of all predictions will be chosen.

Similar to Support Vector Machine(SVM), Random Forest can tackle both classification and regression cases. For prediction, each test data point is passed through every decision tree in the forest. The trees then vote on an outcome and the prediction is produced from a majority vote among the models, resulting in a stronger and more robust single learner. The algorithm is presented in Algorithm 2.

Algorithm 2. Random Forest (RF)

- Step 1: Begin by defining the number of trees to grow, T.
- Step 2: For each tree, randomly select a subset of the training dataset of size n, with replacement. This is called bootstrap aggregating or bagging.
- Step 3: For each subset, randomly select a subset of the features, m, to use as the splitting criteria for each node in the tree.
- Step 4: Grow the decision tree using the selected subset of the training dataset and features.
- Step 5: Repeat steps 2 to 4 for T trees.
- Step 6: To make predictions on new data, pass the data through each decision tree and assign the new data point to the class that is most frequent among the T trees.

K-Nearest Neighbour (KNN)

K-Nearest Neighbour (Yarragunta, 2021) is another classification algorithm. In KNN, for each test data point, we would be looking at the K nearest training data points and take the most frequently occurring classes and assign those to the test data. Therefore, K represents the number of training

data points lying in proximity to the test data point which we are going to use to find the class. The algorithm is presented in Algorithm 3.

Algorithm 3. K-Nearest Neighbour (KNN)

- Step 1: Begin by defining the value of K, the number of neighbours to consider.
- Step 2: For each data point in the dataset, calculate its distance to the new data point using a distance metric such as Euclidean distance.
- Step 3: Sort the distances in ascending order and select the K nearest neighbours.
- Step 4: Assign the new data point to the class that is most frequent among its K nearest neighbours.
- Step 5: Repeat steps 2 to 4 for each new data point in the test dataset.

Decision Tree (DT)

The Decision Tree algorithm (Bekkar, 2022) is a well-known machine learning model that falls under the category of supervised learning. Moreover, it can be used for both classification and prediction tasks. The basic idea of DT is to use the tree representation to solve the proposed problem in which each leaf node corresponds to a specific class label and columns are represented on the internal nodes of the created tree. The main problem in applying DT is selection of the attribute for the root node in each level. Decision Tree algorithms rely on metrics such as Information Gain and Gini Index to evaluate the quality of splits and determine the best attribute to split on. Information Gain measures the reduction in entropy or disorder in the target variable T by splitting on attribute A, while Gini Index measures the probability of misclassifying a random sample in a subset. A lower value of these metrics indicates a better split and a more homogeneous subset. By choosing the best attribute to split on at each node, Decision Tree algorithms can build a tree that provides accurate predictions on the target variable. The procedure is given in Algorithm 4.

Algorithm 4. Decision Tree (DT)

- Step 1: Begin by checking the stopping criteria, such as the maximum depth of the tree or the minimum number of samples in a leaf node.
- Step 2: If the stopping criteria are met, create a leaf node and assign it the most frequent class in the subset.
- Step 3: If the stopping criteria are not met, evaluate each attribute to find the best one to split on. Use Information Gain or Gini Index as the evaluation metric.
- Step 4: Split the subset into two or more subsets based on the chosen attribute.
- Step 5: Recursively repeat steps 1 to 4 for each subset until the stopping criteria are met.
- Step 6: Once the tree is built, use it to make predictions on new data by traversing the tree from the root to a leaf node based on the values of the attributes.
- Step 7: Assign the new data to the class of the leaf node.
- Step 8: Repeat steps 6 and 7 for each new data point.

Discussion and Results

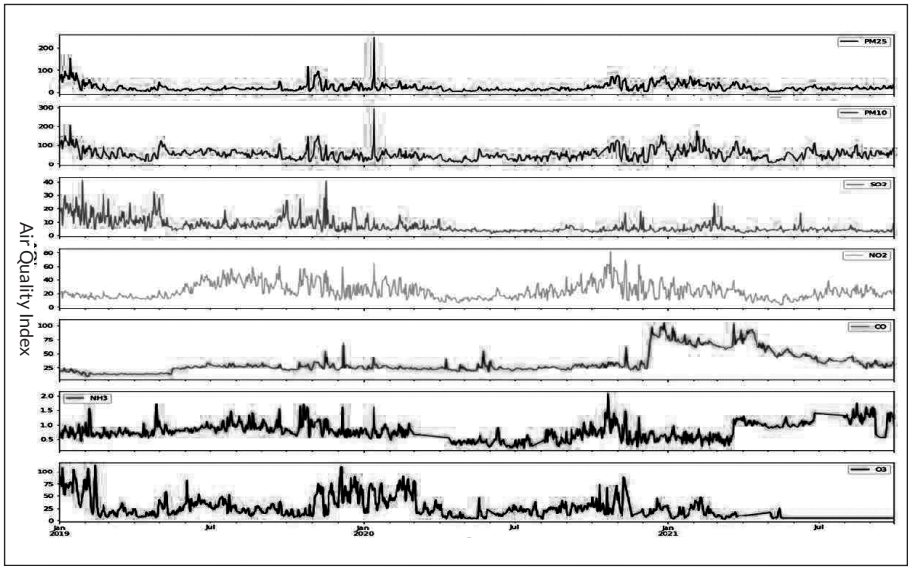
The air pollution quality analysis was done for three stations, namely, Royapuram, Perungudi and Kodungaiyur. The datasets were collected from the TNPCB for the years 2019 to 2021. Each station-wise analysis is presented in this section.

Royapuram

Royapuram is a locality in the northern part of the city of Chennai, Tamil Nadu. For several decades, residents in Royapuram Assembly constituency have been suffering the effects of inhaling limestone, iron ore and coal dust. Transportation of these dusty cargoes in open trucks from and to Chennai port had exposed them to serious health hazards. Surveys revealed that incidences of respiratory diseases and chest infections were higher in the area due to heavy vehicle traffic and the dust. Residents say that port authorities stock iron ore in large heaps in open space in the harbour and that sea breeze carries dust particles to their homes. The Institute of Thoracic Medicine conducted a camp ahead of World Chronic Obstructive Pulmonary Disease (COPD) Day. A total of 412 persons were screened at the camp and the investigations included spirometry, blood and sputum tests and x-ray. The sporadic surveys found that the incidence of COPD and asthma was high in the area.

Earlier, COPD was thought to be caused by smoking but studies have found that indoor stoves and outdoor pollution caused by automobiles was a major cause. Figure 2 depicts the variations in the measured level of pollutants for the years 2019 to 2021. There are high peaks as well as very low and moderate peaks in some of the observations.

Figure 2: Royapuram: Major Pollutants Trend for 2019–2021

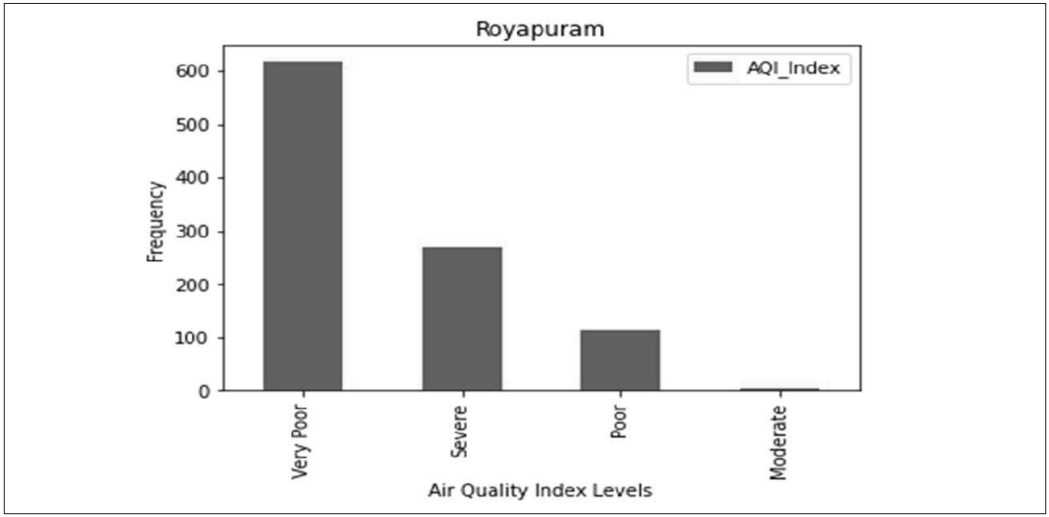


Source: TNPCB (https://tnpcb.gov.in/)

From Figure 2 we observe that during January 2020, the levels of PM2.5 and PM10 are above the 200 AQI level, i.e severe. The SO2 levels appear to be moderate post 2020. NO2 ranges moderate to poor in these years. CO ranges 75 to 100 AQI during the month of December 2019 to June 2021. NH3 values are moderate during these years and O3 levels have a lot of variation between January 2019 and January 2020.

The AQI index classification for Royapuram station is represented in Figure 3.

Figure 3: AQI Index Classification for Royapuram



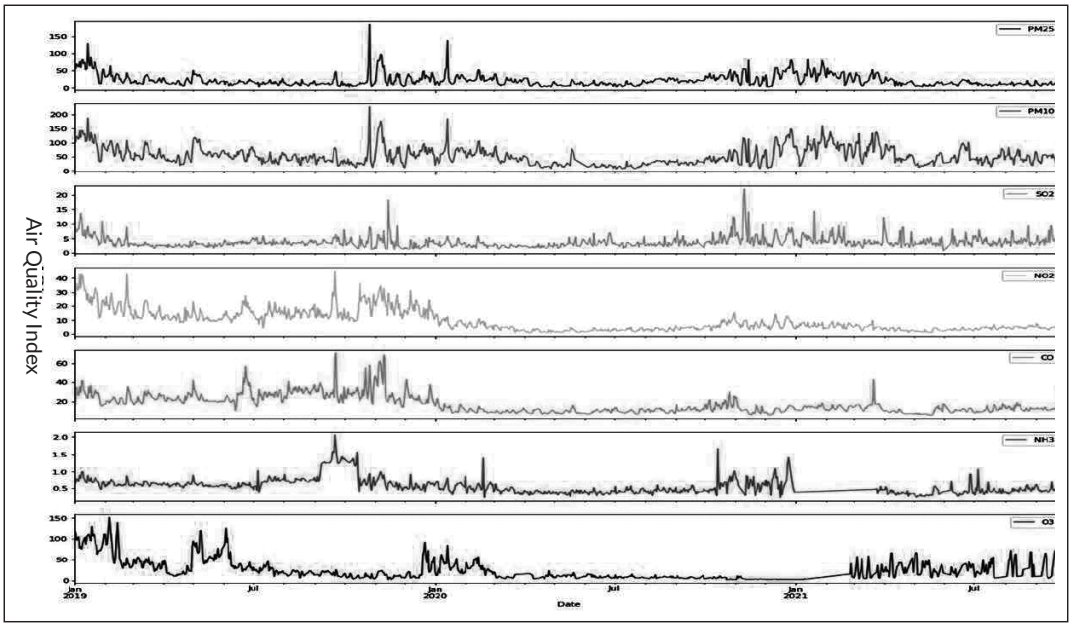
Source: TNPCB (https://tnpcb.gov.in/)

Perungudi

Perungudi is a neighbourhood of Chennai in the state of Tamil Nadu, India. It is situated about 10 kilometres (6.2 miles) south of Adyar. It is bordered on two sides by the Old Mahabalipuram Road and the Perungudi lake. With smoke billowing from the Perungudi dump, levels of various pollutants, including PM10 and ozone in the ambient air have climbed up over the years. Several families living within a one kilometre radius of the Perungudi dump yard, say the toxic pollution was causing respiratory issues, throat irritation and cough.

The emission of methane gas has possibly caused the dump yard fire because of the dumping of large quantities of garden waste. Wind direction, temperature and humidity also contributed to the spread of the fire, so burning in the dump yard was a recurring problem. The air was thick with ash and smoke that causes suffocation and breathing difficulties. Burnt particulate matter from the dump yard was suspended in the air around most of the residential neighbourhoods in the area over the past few years. The pollution levels near the dumpsite, levels of PM10 (particulate matter—dust —of size less than 10 microns), had gone up alarmingly when the fire raged. The levels, however, came down as the fire was put out. The elemental analysis of PM2.5 samples collected at the dumpsite indicated higher concentrations of certain toxic elements such as arsenic, cadmium, copper, chromium, nickel, lead and zinc during winter. The results indicated that the average concentrations of toxic heavy metals were found to be much higher than such metals observed in ambient PM2.5 emissions. Figure 4 depicts the variations in the measured levels of pollutants for the years 2019 to 2021.

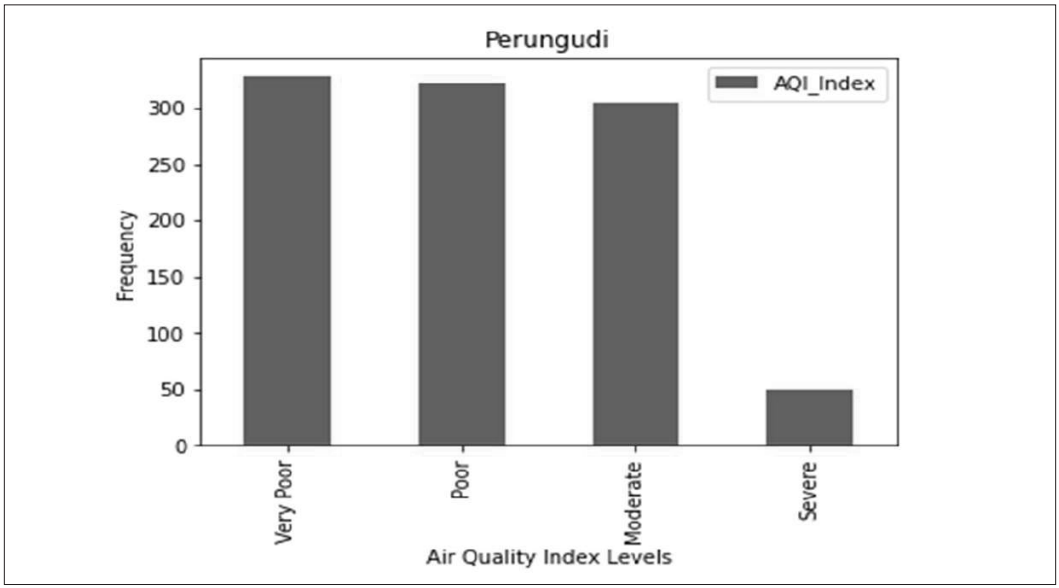
Figure 4: Perungudi: Major Pollutants Trend for 2019–2021



Source: TNPCB (https://tnpcb.gov.in/)

The AQI index classification for Perungudi station is represented in Figure 5.

Figure 5: AQI Index Classification for Perungudi



Source: TNPCB (https://tnpcb.gov.in/)

From Figure 4 we observe that during November 2019 to December 2019, the levels of PM2.5 and PM10 are above the 200 AQI level, i.e. severe. The SO2 levels appear to be moderate and there were a few peaks during 2019. NO2 was high during the month of August 2019 and January 2020. CO ranges were moderate during the months of September 2019 and December 2019. NH3 values were moderate during these years. O3 levels had a lot of variation between January 2019 and July 2019, and post March 2021.

Kodungaiyur

Kodungaiyur is a residential neighbourhood in the northern part of the city of Chennai, Tamil Nadu, India. It comes under Perambur Taluk of the Chennai District. Several studies and analyses have underlined the huge toll on human health and well-being that the neighbourhood dump yard takes. Comptroller and Auditor General of India(CAG) used the Human Rights Impact Assessment (HRIA) concept to assess the impact of waste management practices on the citizens in Kodungaiyur.

CAG has completed the initial phases of a planned two-year long survey among four focus groups in Kodungaiyur— residents, scrap shop workers, conservancy workers, and informal waste pickers. The aim was to understand issues relating to health, society, property and environment arising from the waste dumping. Self-reported data on the symptoms faced due to garbage accumulation was collected from 66 participants from all strata of society. The findings were:

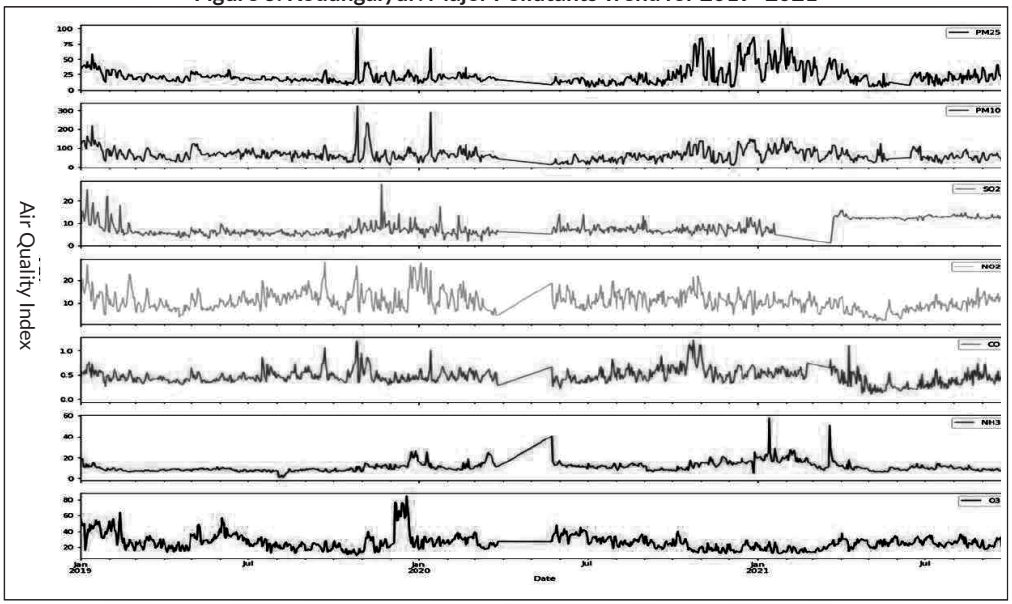
- 31.7 per cent of the residents experienced symptoms that affect the skeletal and muscular system. Symptoms included severe to mild pain in the limbs, stiff joints and back pain.
- 32.8 per cent respondents faced common respiratory issues such as persistent cold and cough, sneezing, wheezing and breathing difficulties. Respondents said these symptoms were due to the burning of garbage at Kodungaiyur.
- 8.5 per cent of the respondents experienced headache and sleeplessness.
- 7.48 per cent had eye infections.
- 7.2 per cent of the respondents complained of skin infections, itchiness, flaky skin and burning sensation.

Samples of air, water, and leachate from different locations in Kodungaiyur tested positive for the presence of heavy metals and volatile organic compounds. 'The permissible limits set by the Bureau of Indian Standards (BIS) and the US Environmental Protection Agency (USEPA) for certain metals have been exceeded for some heavy metals,' says the CAG report.

The AQI index classification for Kodungaiyur station is represented in Figure 7.

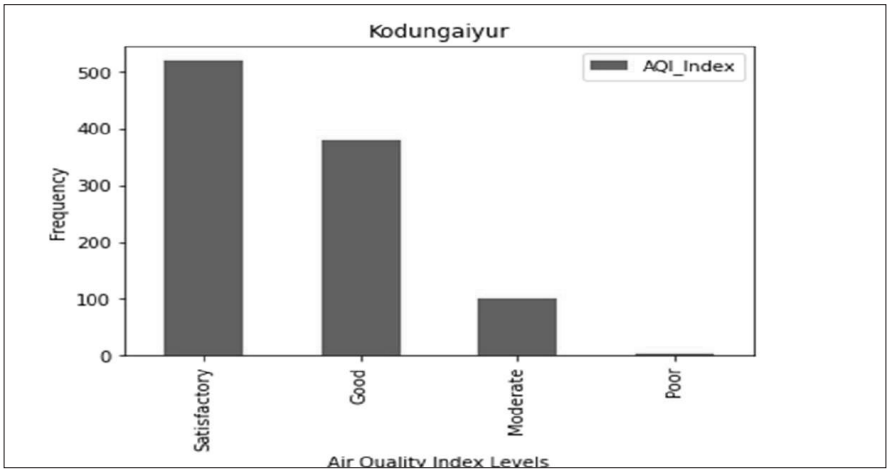
From Figure 6 we observe that from November 2019 to January 2020 the levels of PM2.5 and PM10 were high and continued to be have many variations between November 2020 to February 2021. The SO2 levels appear to be moderate and there were a few peaks during 2019. NO2 and CO levels were moderate on average during these years as were NH3 values. O3 levels had a lot of variations between December 2019 January 2020.

Figure 6: Kodungaiyur: Major Pollutants Trend for 2019–2021



Source: TNPCB (https://tnpcb.gov.in/)

Figure 7: AQI Index Classification for Kodungaiyur



Source: TNPCB (https://tnpcb.gov.in/)

Classification Result Analysis

For air pollution classification the Random Forest (RF), K-Nearest Neighbour (KNN) and Decision Tree(DT) machine learning methods are employed in this work. The datasets were taken for the years 2019 to 2021 covering the zones of Kodungaiyur, Perungudi and Royapuram. The air pollution parameters taken for the study were PM10, PM2.5, SO2, NO2, CO, NH3 and O3.

The classification results of air pollution levels for the various stations are presented in Table 2. Analysis of the results shows an accuracy of 100 per cent, 100 per cent and 98 per cent pertaining to the classifiers Random Forest, Decision Tree and KNN, respectively.

Table 2: Classifier Results for Kodungaiyur, Royapuram and Perungudi

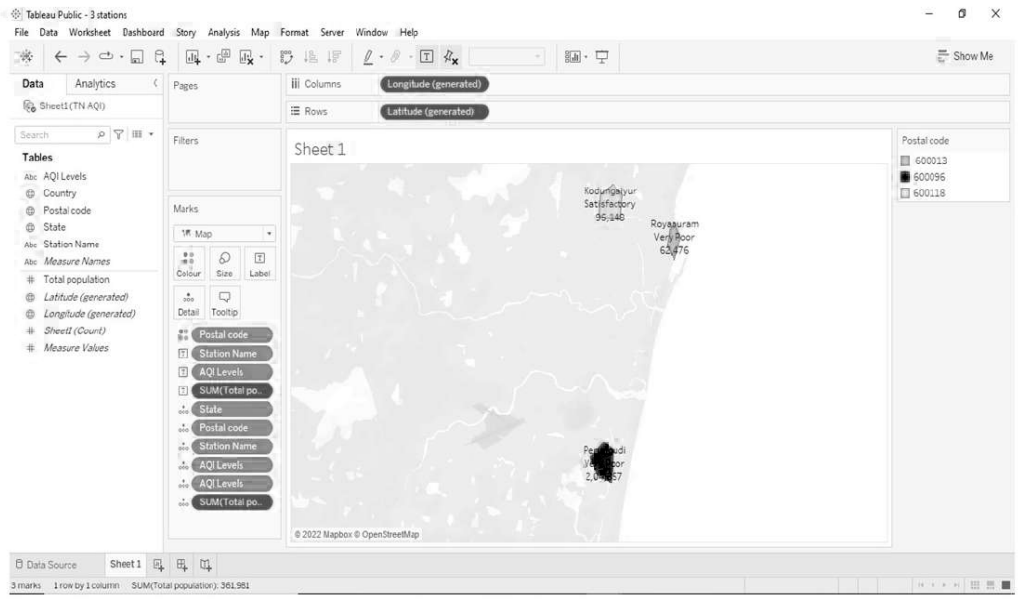
Station Name	Random Forest			Decision Tree			KNN		
	Accuracy	F1 Score	Precision	Accuracy	F1 Score	Precision	Accuracy	F1 Score	Precision
Kodungaiyur	100	100	100	100	100	100	99	99	98
Royapuram	100	100	100	100	100	100	97	100	99
Perungudi	100	100	100	100	100	100	100	100	100

Source: Calculated by Authors

The classification levels of pollutions on a station-wise analysis are depicted in Figure 8 and 9a-9c.

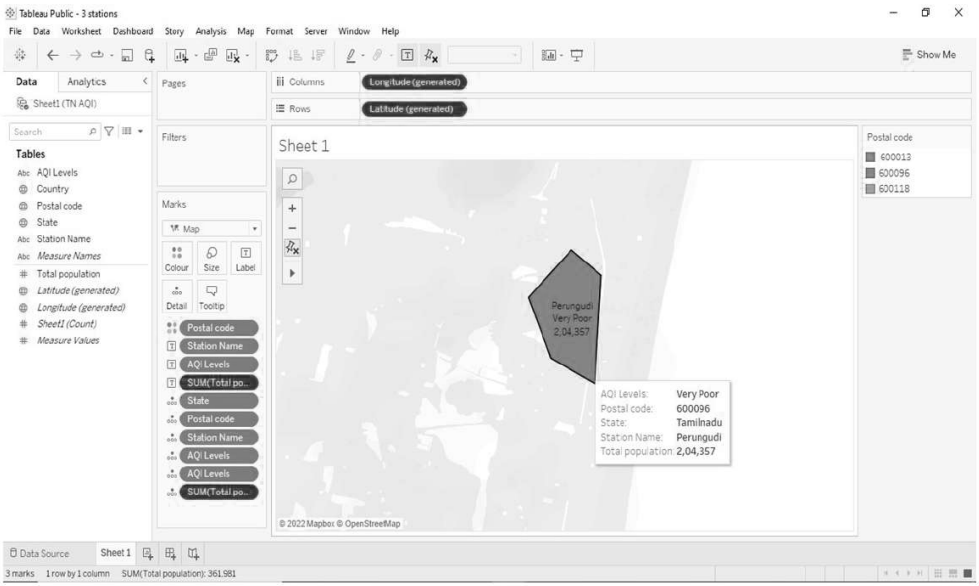
The pollutant results analysis is shown in Figure 10.

Figure 8: Classification of Pollutions in Station-wise Analysis



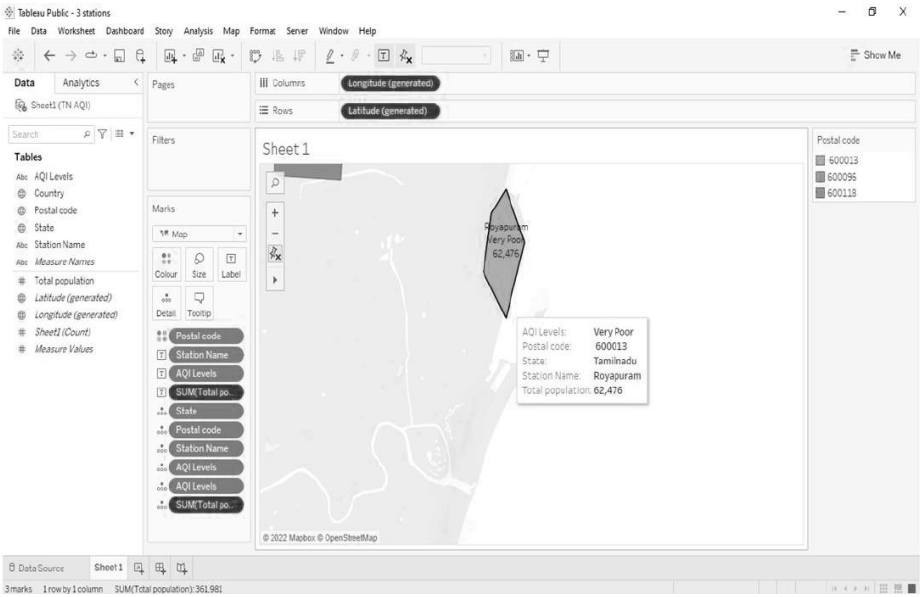
Source: Calculated by Authors

Figure 9a: Analysis of Pollutants – Perungudi



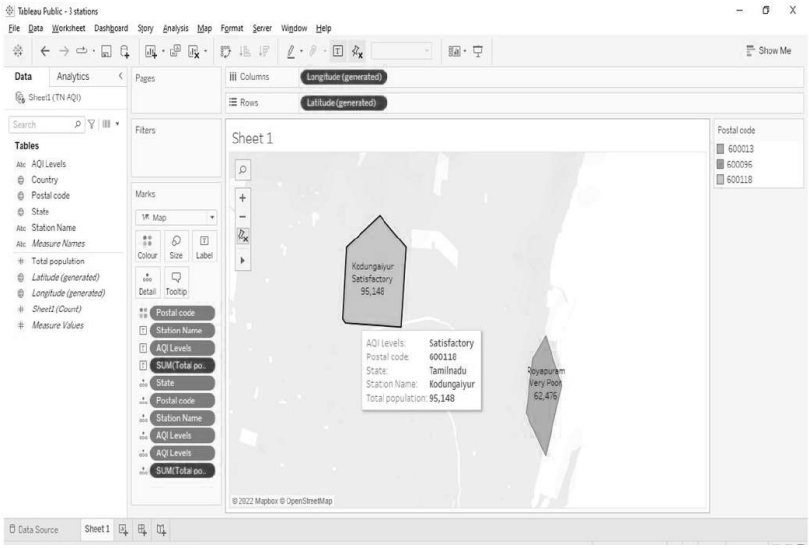
Source: Calculated by Authors

Figure 9b: Analysis of Pollutants – Royapuram



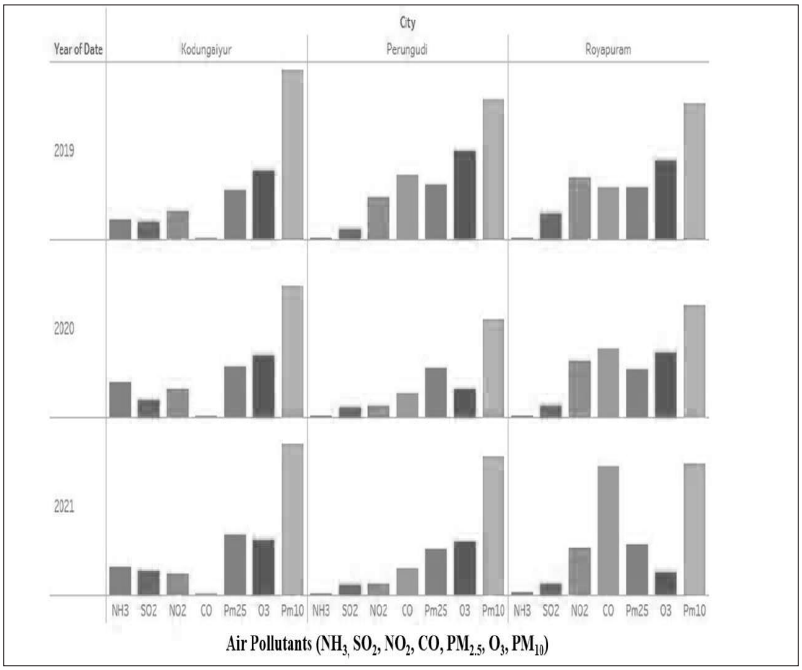
Source: Calculated by Authors

Figure 9c: Analysis of Pollutants – Kodungaiyur



Source: Calculated by Authors

Figure 10: Pollutant Wise Analysis – Pollutant Vs Year



Note: Air Pollutants (NH3, SO2, NO2, CO, PM2.5, O3, PM10)

Source: TNPCB (https://tnpcb.gov.in/)

Conclusion

Air pollution is a serious issue that can cause both acute and chronic health problems in living beings. To address this issue, innovative ideas must be implemented to control and maintain pollutant levels within standard limits. In this study, data from the TNPCB in India for a period of three years was collected to analyse air pollution levels in three regions: Royapuram, Kodungaiyur, and Perungudi of Chennai city in Tamil Nadu. To classify and analyse the air pollution data, the study employed Machine Learning methods such as Random Forest (RF), K-Nearest Neighbour (KNN), and Decision Tree (DT). The analysis results demonstrated a high level of accuracy, with both the Random Forest and Decision Tree methods achieving a 100 per cent accuracy rate, and the KNN method achieving a 98% accuracy rate. This study highlights the potential of Machine Learning methods in accurately analysing and managing air pollution levels, ultimately leading to improved public health and a cleaner environment.

It was realised that there are several preventive measures and policies that can be implemented to control air pollution in the regions of Royapuram, Kodungaiyur, and Perungudi. Here are some critical suggestions:

- Encourage the use of public transportation, car pooling, and cycling to reduce vehicular emissions.
- Implement policies to promote the use of cleaner energy sources, such as solar and wind power.
- Promote green spaces and planting trees to increase the natural air-cleaning abilities of the environment.
- Enforce strict regulations on industries and factories to control their emissions and encourage the use of cleaner technologies.
- Educate the public on the importance of proper waste disposal, including segregation of waste and recycling.
- Introduce and enforce building codes and standards for energy efficiency and green building practices.
- Provide financial incentives for businesses and individuals who adopt cleaner technologies and practices.
- Regularly monitor air quality and make the data publicly available to increase awareness and encourage action.

By implementing these preventive measures and policies, it is possible to reduce air pollution in the regions of Royapuram, Kodungaiyur, and Perungudi. However, it is crucial to have the cooperation and support of the government, industries, and the public to ensure their success.

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Declaration of Conflicting Interest
The Authors hereby declares that there is no conflict of interest.

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